

Contents

1	Documentation conventions	57
1.1	List of abbreviations for registers	57
1.2	Glossary	58
1.3	Peripheral availability	58
2	Memory and bus architecture	59
2.1	System architecture	59
2.1.1	I-bus	62
2.1.2	D-bus	62
2.1.3	S-bus	62
2.1.4	DMA memory bus	63
2.1.5	DMA peripheral bus	63
2.1.6	Ethernet DMA bus	63
2.1.7	USB OTG HS DMA bus	63
2.1.8	LCD-TFT controller DMA bus	63
2.1.9	DMA2D bus	63
2.1.10	BusMatrix	63
2.1.11	AHB/APB bridges (APB)	64
2.2	Memory organization	64
2.3	Memory map	64
2.3.1	Embedded SRAM	68
2.3.2	Flash memory overview	68
2.3.3	Bit banding	68
2.4	Boot configuration	69
3	Embedded flash memory interface	73
3.1	Introduction	73
3.2	Main features	73
3.3	Embedded flash memory in STM32F405xx/07xx and STM32F415xx/17xx	74
3.4	Embedded flash memory in STM32F42xxx and STM32F43xxx	76
3.5	Read interface	80
3.5.1	Relation between CPU clock frequency and flash memory read time	80

3.5.2	Adaptive real-time memory accelerator (ART Accelerator™)	82
3.6	Erase and program operations	84
3.6.1	Unlocking the Flash control register	84
3.6.2	Program/erase parallelism	85
3.6.3	Erase	85
3.6.4	Programming	86
3.6.5	Read-while-write (RWW)	87
3.6.6	Interrupts	88
3.7	Option bytes	88
3.7.1	Description of user option bytes	88
3.7.2	Programming user option bytes	92
3.7.3	Read protection (RDP)	93
3.7.4	Write protections	94
3.7.5	Proprietary code readout protection (PCROP)	95
3.8	One-time programmable bytes	97
3.9	Flash interface registers	98
3.9.1	Flash access control register (FLASH_ACR) for STM32F405xx/07xx and STM32F415xx/17xx	98
3.9.2	Flash access control register (FLASH_ACR) for STM32F42xxx and STM32F43xxx	99
3.9.3	Flash key register (FLASH_KEYR)	100
3.9.4	Flash option key register (FLASH_OPTKEYR)	100
3.9.5	Flash status register (FLASH_SR) for STM32F405xx/07xx and STM32F415xx/17xx	101
3.9.6	Flash status register (FLASH_SR) for STM32F42xxx and STM32F43xxx	102
3.9.7	Flash control register (FLASH_CR) for STM32F405xx/07xx and STM32F415xx/17xx	103
3.9.8	Flash control register (FLASH_CR) for STM32F42xxx and STM32F43xxx	105
3.9.9	Flash option control register (FLASH_OPTCR) for STM32F405xx/07xx and STM32F415xx/17xx	106
3.9.10	Flash option control register (FLASH_OPTCR) for STM32F42xxx and STM32F43xxx	108
3.9.11	Flash option control register (FLASH_OPTCR1) for STM32F42xxx and STM32F43xxx	110
3.9.12	Flash interface register map	111
4	CRC calculation unit	113

4.1	CRC introduction	113
4.2	CRC main features	113
4.3	CRC functional description	114
4.4	CRC registers	114
4.4.1	Data register (CRC_DR)	114
4.4.2	Independent data register (CRC_IDR)	114
4.4.3	Control register (CRC_CR)	115
4.4.4	CRC register map	115
5	Power controller (PWR)	116
5.1	Power supplies	116
5.1.1	Independent A/D converter supply and reference voltage	117
5.1.2	Battery backup domain	118
5.1.3	Voltage regulator for STM32F405xx/07xx and STM32F415xx/17xx ..	120
5.1.4	Voltage regulator for STM32F42xxx and STM32F43xxx	121
5.2	Power supply supervisor	124
5.2.1	Power-on reset (POR)/power-down reset (PDR)	124
5.2.2	Brownout reset (BOR)	125
5.2.3	Programmable voltage detector (PVD)	125
5.3	Low-power modes	126
5.3.1	Slowing down system clocks	128
5.3.2	Peripheral clock gating	128
5.3.3	Sleep mode	129
5.3.4	Stop mode (STM32F405xx/07xx and STM32F415xx/17xx)	130
5.3.5	Stop mode (STM32F42xxx and STM32F43xxx)	133
5.3.6	Standby mode	136
5.3.7	Programming the RTC alternate functions to wake up the device from the Stop and Standby modes	138
5.4	Power control registers (STM32F405xx/07xx and STM32F415xx/17xx) ..	141
5.4.1	PWR power control register (PWR_CR) for STM32F405xx/07xx and STM32F415xx/17xx	141
5.4.2	PWR power control/status register (PWR_CSR) for STM32F405xx/07xx and STM32F415xx/17xx	142
5.5	Power control registers (STM32F42xxx and STM32F43xxx)	145
5.5.1	PWR power control register (PWR_CR) for STM32F42xxx and STM32F43xxx	145
5.5.2	PWR power control/status register (PWR_CSR) for STM32F42xxx and STM32F43xxx	148

5.6	PWR register map	150
6	Reset and clock control for STM32F42xxx and STM32F43xxx (RCC)	151
6.1	Reset	151
6.1.1	System reset	151
6.1.2	Power reset	151
6.1.3	Backup domain reset	152
6.2	Clocks	152
6.2.1	HSE clock	155
6.2.2	HSI clock	156
6.2.3	PLL configuration	156
6.2.4	LSE clock	157
6.2.5	LSI clock	157
6.2.6	System clock (SYSCLK) selection	157
6.2.7	Clock security system (CSS)	158
6.2.8	RTC/AWU clock	158
6.2.9	Watchdog clock	159
6.2.10	Clock-out capability	159
6.2.11	Internal/external clock measurement using TIM5/TIM11	159
6.3	RCC registers	162
6.3.1	RCC clock control register (RCC_CR)	162
6.3.2	RCC PLL configuration register (RCC_PLLCFGR)	164
6.3.3	RCC clock configuration register (RCC_CFGR)	166
6.3.4	RCC clock interrupt register (RCC_CIR)	168
6.3.5	RCC AHB1 peripheral reset register (RCC_AHB1RSTR)	171
6.3.6	RCC AHB2 peripheral reset register (RCC_AHB2RSTR)	174
6.3.7	RCC AHB3 peripheral reset register (RCC_AHB3RSTR)	175
6.3.8	RCC APB1 peripheral reset register (RCC_APB1RSTR)	175
6.3.9	RCC APB2 peripheral reset register (RCC_APB2RSTR)	179
6.3.10	RCC AHB1 peripheral clock register (RCC_AHB1ENR)	181
6.3.11	RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)	183
6.3.12	RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)	184
6.3.13	RCC APB1 peripheral clock enable register (RCC_APB1ENR)	184
6.3.14	RCC APB2 peripheral clock enable register (RCC_APB2ENR)	188
6.3.15	RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR)	190

6.3.16	RCC AHB2 peripheral clock enable in low power mode register (RCC_AHB2LPENR)	193
6.3.17	RCC AHB3 peripheral clock enable in low power mode register (RCC_AHB3LPENR)	194
6.3.18	RCC APB1 peripheral clock enable in low power mode register (RCC_APB1LPENR)	194
6.3.19	RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR)	198
6.3.20	RCC Backup domain control register (RCC_BDCR)	200
6.3.21	RCC clock control & status register (RCC_CSR)	201
6.3.22	RCC spread spectrum clock generation register (RCC_SSCGR)	203
6.3.23	RCC PLLI2S configuration register (RCC_PLLI2SCFGR)	204
6.3.24	RCC PLL configuration register (RCC_PLLSAICFGR)	207
6.3.25	RCC Dedicated Clock Configuration Register (RCC_DCKCFGR) ...	208
6.3.26	RCC register map	211

7	Reset and clock control for STM32F405xx/07xx and STM32F415xx/17xx(RCC)	214
7.1	Reset	214
7.1.1	System reset	214
7.1.2	Power reset	215
7.1.3	Backup domain reset	215
7.2	Clocks	216
7.2.1	HSE clock	218
7.2.2	HSI clock	219
7.2.3	PLL configuration	220
7.2.4	LSE clock	220
7.2.5	LSI clock	221
7.2.6	System clock (SYSCLK) selection	221
7.2.7	Clock security system (CSS)	221
7.2.8	RTC/AWU clock	222
7.2.9	Watchdog clock	222
7.2.10	Clock-out capability	223
7.2.11	Internal/external clock measurement using TIM5/TIM11	223
7.3	RCC registers	225
7.3.1	RCC clock control register (RCC_CR)	225
7.3.2	RCC PLL configuration register (RCC_PLLCFGR)	227
7.3.3	RCC clock configuration register (RCC_CFGR)	229

7.3.4	RCC clock interrupt register (RCC_CIR)	231
7.3.5	RCC AHB1 peripheral reset register (RCC_AHB1RSTR)	234
7.3.6	RCC AHB2 peripheral reset register (RCC_AHB2RSTR)	237
7.3.7	RCC AHB3 peripheral reset register (RCC_AHB3RSTR)	238
7.3.8	RCC APB1 peripheral reset register (RCC_APB1RSTR)	238
7.3.9	RCC APB2 peripheral reset register (RCC_APB2RSTR)	241
7.3.10	RCC AHB1 peripheral clock enable register (RCC_AHB1ENR)	243
7.3.11	RCC AHB2 peripheral clock enable register (RCC_AHB2ENR)	245
7.3.12	RCC AHB3 peripheral clock enable register (RCC_AHB3ENR)	246
7.3.13	RCC APB1 peripheral clock enable register (RCC_APB1ENR)	246
7.3.14	RCC APB2 peripheral clock enable register (RCC_APB2ENR)	249
7.3.15	RCC AHB1 peripheral clock enable in low power mode register (RCC_AHB1LPENR)	251
7.3.16	RCC AHB2 peripheral clock enable in low power mode register (RCC_AHB2LPENR)	253
7.3.17	RCC AHB3 peripheral clock enable in low power mode register (RCC_AHB3LPENR)	254
7.3.18	RCC APB1 peripheral clock enable in low power mode register (RCC_APB1LPENR)	255
7.3.19	RCC APB2 peripheral clock enabled in low power mode register (RCC_APB2LPENR)	258
7.3.20	RCC Backup domain control register (RCC_BDCR)	260
7.3.21	RCC clock control & status register (RCC_CSR)	261
7.3.22	RCC spread spectrum clock generation register (RCC_SSCGR)	263
7.3.23	RCC PLLI2S configuration register (RCC_PLLI2SCFGR)	264
7.3.24	RCC register map	266
8	General-purpose I/Os (GPIO)	269
8.1	GPIO introduction	269
8.2	GPIO main features	269
8.3	GPIO functional description	269
8.3.1	General-purpose I/O (GPIO)	271
8.3.2	I/O pin multiplexer and mapping	272
8.3.3	I/O port control registers	276
8.3.4	I/O port data registers	276
8.3.5	I/O data bitwise handling	276
8.3.6	GPIO locking mechanism	276
8.3.7	I/O alternate function input/output	277

8.3.8	External interrupt/wake-up lines	277
8.3.9	Input configuration	277
8.3.10	Output configuration	278
8.3.11	Alternate function configuration	279
8.3.12	Analog configuration	280
8.3.13	Using the OSC32_IN/OSC32_OUT pins as GPIO PC14/PC15 port pins	280
8.3.14	Using the OSC_IN/OSC_OUT pins as GPIO PH0/PH1 port pins	280
8.3.15	Selection of RTC_AF1 and RTC_AF2 alternate functions	281
8.4	GPIO registers	283
8.4.1	GPIO port mode register (GPIOx_MODER) (x = A..I/J/K)	283
8.4.2	GPIO port output type register (GPIOx_OTYPER) (x = A..I/J/K)	283
8.4.3	GPIO port output speed register (GPIOx_OSPEEDR) (x = A..I/J/K)	284
8.4.4	GPIO port pull-up/pull-down register (GPIOx_PUPDR) (x = A..I/J/K)	284
8.4.5	GPIO port input data register (GPIOx_IDR) (x = A..I/J/K)	285
8.4.6	GPIO port output data register (GPIOx_ODR) (x = A..I/J/K)	285
8.4.7	GPIO port bit set/reset register (GPIOx_BSRR) (x = A..I/J/K)	286
8.4.8	GPIO port configuration lock register (GPIOx_LCKR) (x = A..I/J/K)	286
8.4.9	GPIO alternate function low register (GPIOx_AFRL) (x = A..I/J/K)	287
8.4.10	GPIO alternate function high register (GPIOx_AFRH) (x = A..I/J)	288
8.4.11	GPIO register map	289
9	System configuration controller (SYSCFG)	291
9.1	I/O compensation cell	291
9.2	SYSCFG registers for STM32F405xx/07xx and STM32F415xx/17xx	291
9.2.1	SYSCFG memory remap register (SYSCFG_MEMRMP)	291
9.2.2	SYSCFG peripheral mode configuration register (SYSCFG_PMC)	292
9.2.3	SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)	293
9.2.4	SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)	293
9.2.5	SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)	294
9.2.6	SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)	295

9.2.7	Compensation cell control register (SYSCFG_CMPCR)	295
9.2.8	SYSCFG register maps for STM32F405xx/07xx and STM32F415xx/17xx	296
9.3	SYSCFG registers for STM32F42xxx and STM32F43xxx	296
9.3.1	SYSCFG memory remap register (SYSCFG_MEMRMP)	296
9.3.2	SYSCFG peripheral mode configuration register (SYSCFG_PMC)	298
9.3.3	SYSCFG external interrupt configuration register 1 (SYSCFG_EXTICR1)	299
9.3.4	SYSCFG external interrupt configuration register 2 (SYSCFG_EXTICR2)	300
9.3.5	SYSCFG external interrupt configuration register 3 (SYSCFG_EXTICR3)	300
9.3.6	SYSCFG external interrupt configuration register 4 (SYSCFG_EXTICR4)	301
9.3.7	Compensation cell control register (SYSCFG_CMPCR)	302
9.3.8	SYSCFG register maps for STM32F42xxx and STM32F43xxx	303
10	DMA controller (DMA)	304
10.1	DMA introduction	304
10.2	DMA main features	304
10.3	DMA functional description	306
10.3.1	General description	306
10.3.2	DMA transactions	308
10.3.3	Channel selection	309
10.3.4	Arbiter	310
10.3.5	DMA streams	311
10.3.6	Source, destination and transfer modes	311
10.3.7	Pointer incrementation	314
10.3.8	Circular mode	315
10.3.9	Double buffer mode	315
10.3.10	Programmable data width, packing/unpacking, endianness	316
10.3.11	Single and burst transfers	318
10.3.12	FIFO	319
10.3.13	DMA transfer completion	321
10.3.14	DMA transfer suspension	322
10.3.15	Flow controller	322
10.3.16	Summary of the possible DMA configurations	324
10.3.17	Stream configuration procedure	324

10.3.18	Error management	325
10.4	DMA interrupts	326
10.5	DMA registers	327
10.5.1	DMA low interrupt status register (DMA_LISR)	327
10.5.2	DMA high interrupt status register (DMA_HISR)	328
10.5.3	DMA low interrupt flag clear register (DMA_LIFCR)	329
10.5.4	DMA high interrupt flag clear register (DMA_HIFCR)	329
10.5.5	DMA stream x configuration register (DMA_SxCR) (x = 0..7)	330
10.5.6	DMA stream x number of data register (DMA_SxNDTR) (x = 0..7)	333
10.5.7	DMA stream x peripheral address register (DMA_SxPAR) (x = 0..7)	333
10.5.8	DMA stream x memory 0 address register (DMA_SxM0AR) (x = 0..7)	334
10.5.9	DMA stream x memory 1 address register (DMA_SxM1AR) (x = 0..7)	334
10.5.10	DMA stream x FIFO control register (DMA_SxFCR) (x = 0..7)	335
10.5.11	DMA register map	337
11	Chrom-Art Accelerator™ controller (DMA2D)	341
11.1	DMA2D introduction	341
11.2	DMA2D main features	342
11.3	DMA2D functional description	342
11.3.1	General description	342
11.3.2	DMA2D control	343
11.3.3	DMA2D foreground and background FIFOs	343
11.3.4	DMA2D foreground and background pixel format converter (PFC)	344
11.3.5	DMA2D foreground and background CLUT interface	346
11.3.6	DMA2D blender	347
11.3.7	DMA2D output PFC	347
11.3.8	DMA2D output FIFO	348
11.3.9	DMA2D AHB master port timer	348
11.3.10	DMA2D transactions	349
11.3.11	DMA2D configuration	349
11.3.12	DMA2D transfer control (start, suspend, abort and completion)	352
11.3.13	Watermark	353
11.3.14	Error management	353
11.3.15	AHB dead time	353
11.4	DMA2D interrupts	353
11.5	DMA2D registers	354

11.5.1	DMA2D control register (DMA2D_CR)	354
11.5.2	DMA2D Interrupt Status Register (DMA2D_ISR)	356
11.5.3	DMA2D interrupt flag clear register (DMA2D_IFCR)	357
11.5.4	DMA2D foreground memory address register (DMA2D_FGMAR) ...	358
11.5.5	DMA2D foreground offset register (DMA2D_FGOR)	358
11.5.6	DMA2D background memory address register (DMA2D_BGMAR) ..	359
11.5.7	DMA2D background offset register (DMA2D_BGOR)	359
11.5.8	DMA2D foreground PFC control register (DMA2D_FGPFCCR)	360
11.5.9	DMA2D foreground color register (DMA2D_FGCOLOR)	362
11.5.10	DMA2D background PFC control register (DMA2D_BGPFCCR)	363
11.5.11	DMA2D background color register (DMA2D_BGCOLOR)	365
11.5.12	DMA2D foreground CLUT memory address register (DMA2D_FGCMAR)	365
11.5.13	DMA2D background CLUT memory address register (DMA2D_BGCMAR)	366
11.5.14	DMA2D output PFC control register (DMA2D_OPFCCR)	366
11.5.15	DMA2D output color register (DMA2D_OCOLOR)	367
11.5.16	DMA2D output memory address register (DMA2D_OMAR)	368
11.5.17	DMA2D output offset register (DMA2D_OOR)	369
11.5.18	DMA2D number of line register (DMA2D_NLR)	369
11.5.19	DMA2D line watermark register (DMA2D_LWR)	370
11.5.20	DMA2D AHB master timer configuration register (DMA2D_AMTCR) .	370
11.5.21	DMA2D register map	371
12	Interrupts and events	373
12.1	Nested vectored interrupt controller (NVIC)	373
12.1.1	NVIC features	373
12.1.2	SysTick calibration value register	373
12.1.3	Interrupt and exception vectors	373
12.2	External interrupt/event controller (EXTI)	373
12.2.1	EXTI main features	381
12.2.2	EXTI block diagram	382
12.2.3	Wake-up event management	382
12.2.4	Functional description	382
12.2.5	External interrupt/event line mapping	384
12.3	EXTI registers	386
12.3.1	Interrupt mask register (EXTI_IMR)	386

12.3.2	Event mask register (EXTI_EMR)	386
12.3.3	Rising trigger selection register (EXTI_RTSR)	387
12.3.4	Falling trigger selection register (EXTI_FTSR)	387
12.3.5	Software interrupt event register (EXTI_SWIER)	388
12.3.6	Pending register (EXTI_PR)	388
12.3.7	EXTI register map	389
13	Analog-to-digital converter (ADC)	390
13.1	ADC introduction	390
13.2	ADC main features	390
13.3	ADC functional description	391
13.3.1	ADC on-off control	392
13.3.2	ADC clock	392
13.3.3	Channel selection	392
13.3.4	Single conversion mode	393
13.3.5	Continuous conversion mode	394
13.3.6	Timing diagram	394
13.3.7	Analog watchdog	394
13.3.8	Scan mode	395
13.3.9	Injected channel management	396
13.3.10	Discontinuous mode	397
13.4	Data alignment	398
13.5	Channel-wise programmable sampling time	399
13.6	Conversion on external trigger and trigger polarity	399
13.7	Fast conversion mode	401
13.8	Data management	402
13.8.1	Using the DMA	402
13.8.2	Managing a sequence of conversions without using the DMA	402
13.8.3	Conversions without DMA and without overrun detection	403
13.9	Multi ADC mode	403
13.9.1	Injected simultaneous mode	406
13.9.2	Regular simultaneous mode	407
13.9.3	Interleaved mode	409
13.9.4	Alternate trigger mode	410
13.9.5	Combined regular/injected simultaneous mode	412
13.9.6	Combined regular simultaneous + alternate trigger mode	413

13.10	Temperature sensor	414
13.11	Battery charge monitoring	416
13.12	ADC interrupts	416
13.13	ADC registers	417
13.13.1	ADC status register (ADC_SR)	417
13.13.2	ADC control register 1 (ADC_CR1)	418
13.13.3	ADC control register 2 (ADC_CR2)	420
13.13.4	ADC sample time register 1 (ADC_SMPR1)	422
13.13.5	ADC sample time register 2 (ADC_SMPR2)	422
13.13.6	ADC injected channel data offset register x (ADC_JOFRx) (x=1..4)	423
13.13.7	ADC watchdog higher threshold register (ADC_HTR)	423
13.13.8	ADC watchdog lower threshold register (ADC_LTR)	424
13.13.9	ADC regular sequence register 1 (ADC_SQR1)	424
13.13.10	ADC regular sequence register 2 (ADC_SQR2)	425
13.13.11	ADC regular sequence register 3 (ADC_SQR3)	425
13.13.12	ADC injected sequence register (ADC_JSQR)	426
13.13.13	ADC injected data register x (ADC_JDRx) (x= 1..4)	427
13.13.14	ADC regular data register (ADC_DR)	427
13.13.15	ADC Common status register (ADC_CSR)	428
13.13.16	ADC common control register (ADC_CCR)	429
13.13.17	ADC common regular data register for dual and triple modes (ADC_CDR)	432
13.13.18	ADC register map	432
14	Digital-to-analog converter (DAC)	435
14.1	DAC introduction	435
14.2	DAC main features	435
14.3	DAC functional description	437
14.3.1	DAC channel enable	437
14.3.2	DAC output buffer enable	437
14.3.3	DAC data format	437
14.3.4	DAC conversion	438
14.3.5	DAC output voltage	439
14.3.6	DAC trigger selection	439
14.3.7	DMA request	440
14.3.8	Noise generation	440
14.3.9	Triangle-wave generation	441

14.4	Dual DAC channel conversion	442
14.4.1	Independent trigger without wave generation	443
14.4.2	Independent trigger with single LFSR generation	443
14.4.3	Independent trigger with different LFSR generation	443
14.4.4	Independent trigger with single triangle generation	444
14.4.5	Independent trigger with different triangle generation	444
14.4.6	Simultaneous software start	444
14.4.7	Simultaneous trigger without wave generation	445
14.4.8	Simultaneous trigger with single LFSR generation	445
14.4.9	Simultaneous trigger with different LFSR generation	445
14.4.10	Simultaneous trigger with single triangle generation	446
14.4.11	Simultaneous trigger with different triangle generation	446
14.5	DAC registers	447
14.5.1	DAC control register (DAC_CR)	447
14.5.2	DAC software trigger register (DAC_SWTRIGR)	450
14.5.3	DAC channel1 12-bit right-aligned data holding register (DAC_DHR12R1)	450
14.5.4	DAC channel1 12-bit left aligned data holding register (DAC_DHR12L1)	451
14.5.5	DAC channel1 8-bit right aligned data holding register (DAC_DHR8R1)	451
14.5.6	DAC channel2 12-bit right aligned data holding register (DAC_DHR12R2)	452
14.5.7	DAC channel2 12-bit left aligned data holding register (DAC_DHR12L2)	452
14.5.8	DAC channel2 8-bit right-aligned data holding register (DAC_DHR8R2)	452
14.5.9	Dual DAC 12-bit right-aligned data holding register (DAC_DHR12RD)	453
14.5.10	DUAL DAC 12-bit left aligned data holding register (DAC_DHR12LD)	453
14.5.11	DUAL DAC 8-bit right aligned data holding register (DAC_DHR8RD)	454
14.5.12	DAC channel1 data output register (DAC_DOR1)	454
14.5.13	DAC channel2 data output register (DAC_DOR2)	454
14.5.14	DAC status register (DAC_SR)	455
14.5.15	DAC register map	455
15	Digital camera interface (DCMI)	457

15.1	DCMI introduction	457
15.2	DCMI main features	457
15.3	DCMI pins	457
15.4	DCMI clocks	457
15.5	DCMI functional overview	458
15.5.1	DMA interface	459
15.5.2	DCMI physical interface	459
15.5.3	Synchronization	461
15.5.4	Capture modes	463
15.5.5	Crop feature	465
15.5.6	JPEG format	467
15.5.7	FIFO	467
15.6	Data format description	467
15.6.1	Data formats	467
15.6.2	Monochrome format	468
15.6.3	RGB format	468
15.6.4	YCbCr format	468
15.7	DCMI interrupts	469
15.8	DCMI register description	469
15.8.1	DCMI control register 1 (DCMI_CR)	469
15.8.2	DCMI status register (DCMI_SR)	471
15.8.3	DCMI raw interrupt status register (DCMI_RIS)	472
15.8.4	DCMI interrupt enable register (DCMI_IER)	473
15.8.5	DCMI masked interrupt status register (DCMI_MIS)	474
15.8.6	DCMI interrupt clear register (DCMI_ICR)	475
15.8.7	DCMI embedded synchronization code register (DCMI_ESCR)	475
15.8.8	DCMI embedded synchronization unmask register (DCMI_ESUR)	476
15.8.9	DCMI crop window start (DCMI_CWSTRT)	477
15.8.10	DCMI crop window size (DCMI_CWSIZE)	477
15.8.11	DCMI data register (DCMI_DR)	478
15.8.12	DCMI register map	478
16	LCD-TFT controller (LTDC)	480
16.1	Introduction	480
16.2	LTDC main features	480
16.3	LTDC functional description	481

16.3.1	LTDC block diagram	481
16.3.2	LTDC reset and clocks	481
16.3.3	LCD-TFT pins and signal interface	483
16.4	LTDC programmable parameters	483
16.4.1	LTDC Global configuration parameters	483
16.4.2	Layer programmable parameters	486
16.5	LTDC interrupts	490
16.6	LTDC programming procedure	492
16.7	LTDC registers	493
16.7.1	LTDC Synchronization Size Configuration Register (LTDC_SSCR)	493
16.7.2	LTDC Back Porch Configuration Register (LTDC_BPCR)	493
16.7.3	LTDC Active Width Configuration Register (LTDC_AWCR)	494
16.7.4	LTDC Total Width Configuration Register (LTDC_TWCR)	495
16.7.5	LTDC Global Control Register (LTDC_GCR)	495
16.7.6	LTDC Shadow Reload Configuration Register (LTDC_SRCR)	497
16.7.7	LTDC Background Color Configuration Register (LTDC_BCCR)	497
16.7.8	LTDC Interrupt Enable Register (LTDC_IER)	498
16.7.9	LTDC Interrupt Status Register (LTDC_ISR)	499
16.7.10	LTDC Interrupt Clear Register (LTDC_ICR)	499
16.7.11	LTDC Line Interrupt Position Configuration Register (LTDC_LIPCR)	500
16.7.12	LTDC Current Position Status Register (LTDC_CPSR)	500
16.7.13	LTDC Current Display Status Register (LTDC_CDSR)	501
16.7.14	LTDC Layerx Control Register (LTDC_LxCR) (where x=1..2)	502
16.7.15	LTDC Layerx Window Horizontal Position Configuration Register (LTDC_LxWHPCR) (where x=1..2)	503
16.7.16	LTDC Layerx Window Vertical Position Configuration Register (LTDC_LxWVPCR) (where x=1..2)	504
16.7.17	LTDC Layerx Color Keying Configuration Register (LTDC_LxCKCR) (where x=1..2)	505
16.7.18	LTDC Layerx Pixel Format Configuration Register (LTDC_LxPFCR) (where x=1..2)	505
16.7.19	LTDC Layerx Constant Alpha Configuration Register (LTDC_LxCACR) (where x=1..2)	506
16.7.20	LTDC Layerx Default Color Configuration Register (LTDC_LxDCCR) (where x=1..2)	506
16.7.21	LTDC Layerx Blending Factors Configuration Register (LTDC_LxBFCR) (where x=1..2)	508
16.7.22	LTDC Layerx Color Frame Buffer Address Register (LTDC_LxCFBAR) (where x=1..2)	509

16.7.23	LTDC Layerx Color Frame Buffer Length Register (LTDC_LxCFBLR) (where x=1..2)	509
16.7.24	LTDC Layerx ColorFrame Buffer Line Number Register (LTDC_LxCFBLNR) (where x=1..2)	510
16.7.25	LTDC Layerx CLUT Write Register (LTDC_LxCLUTWR) (where x=1..2)	511
16.7.26	LTDC register map	512
17	Advanced-control timers (TIM1 and TIM8)	515
17.1	TIM1 and TIM8 introduction	515
17.2	TIM1 and TIM8 main features	515
17.3	TIM1 and TIM8 functional description	517
17.3.1	Time-base unit	517
17.3.2	Counter modes	519
17.3.3	Repetition counter	528
17.3.4	Clock selection	531
17.3.5	Capture/compare channels	534
17.3.6	Input capture mode	537
17.3.7	PWM input mode	538
17.3.8	Forced output mode	538
17.3.9	Output compare mode	539
17.3.10	PWM mode	540
17.3.11	Complementary outputs and dead-time insertion	543
17.3.12	Using the break function	545
17.3.13	Clearing the OCxREF signal on an external event	548
17.3.14	6-step PWM generation	549
17.3.15	One-pulse mode	550
17.3.16	Encoder interface mode	551
17.3.17	Timer input XOR function	554
17.3.18	Interfacing with Hall sensors	554
17.3.19	TIMx and external trigger synchronization	556
17.3.20	Timer synchronization	559
17.3.21	Debug mode	559
17.4	TIM1 and TIM8 registers	560
17.4.1	TIM1 and TIM8 control register 1 (TIMx_CR1)	560
17.4.2	TIM1 and TIM8 control register 2 (TIMx_CR2)	561
17.4.3	TIM1 and TIM8 slave mode control register (TIMx_SMCR)	563
17.4.4	TIM1 and TIM8 DMA/interrupt enable register (TIMx_DIER)	565

17.4.5	TIM1 and TIM8 status register (TIMx_SR)	567
17.4.6	TIM1 and TIM8 event generation register (TIMx_EGR)	568
17.4.7	TIM1 and TIM8 capture/compare mode register 1 (TIMx_CCMR1) ..	569
17.4.8	TIM1 and TIM8 capture/compare mode register 2 (TIMx_CCMR2) ..	572
17.4.9	TIM1 and TIM8 capture/compare enable register (TIMx_CCER)	574
17.4.10	TIM1 and TIM8 counter (TIMx_CNT)	577
17.4.11	TIM1 and TIM8 prescaler (TIMx_PSC)	577
17.4.12	TIM1 and TIM8 auto-reload register (TIMx_ARR)	577
17.4.13	TIM1 and TIM8 repetition counter register (TIMx_RCR)	578
17.4.14	TIM1 and TIM8 capture/compare register 1 (TIMx_CCR1)	578
17.4.15	TIM1 and TIM8 capture/compare register 2 (TIMx_CCR2)	579
17.4.16	TIM1 and TIM8 capture/compare register 3 (TIMx_CCR3)	579
17.4.17	TIM1 and TIM8 capture/compare register 4 (TIMx_CCR4)	580
17.4.18	TIM1 and TIM8 break and dead-time register (TIMx_BDTR)	580
17.4.19	TIM1 and TIM8 DMA control register (TIMx_DCR)	582
17.4.20	TIM1 and TIM8 DMA address for full transfer (TIMx_DMAR)	583
17.4.21	TIM1 and TIM8 register map	584
18	General-purpose timers (TIM2 to TIM5)	586
18.1	TIM2 to TIM5 introduction	586
18.2	TIM2 to TIM5 main features	586
18.3	TIM2 to TIM5 functional description	588
18.3.1	Time-base unit	588
18.3.2	Counter modes	589
18.3.3	Clock selection	599
18.3.4	Capture/compare channels	602
18.3.5	Input capture mode	604
18.3.6	PWM input mode	605
18.3.7	Forced output mode	606
18.3.8	Output compare mode	606
18.3.9	PWM mode	607
18.3.10	One-pulse mode	610
18.3.11	Clearing the OCxREF signal on an external event	611
18.3.12	Encoder interface mode	612
18.3.13	Timer input XOR function	615
18.3.14	Timers and external trigger synchronization	615
18.3.15	Timer synchronization	618

18.3.16	Debug mode	623
18.4	TIM2 to TIM5 registers	624
18.4.1	TIMx control register 1 (TIMx_CR1)	624
18.4.2	TIMx control register 2 (TIMx_CR2)	625
18.4.3	TIMx slave mode control register (TIMx_SMCR)	626
18.4.4	TIMx DMA/Interrupt enable register (TIMx_DIER)	629
18.4.5	TIMx status register (TIMx_SR)	630
18.4.6	TIMx event generation register (TIMx_EGR)	631
18.4.7	TIMx capture/compare mode register 1 (TIMx_CCMR1)	632
18.4.8	TIMx capture/compare mode register 2 (TIMx_CCMR2)	635
18.4.9	TIMx capture/compare enable register (TIMx_CCER)	636
18.4.10	TIMx counter (TIMx_CNT)	638
18.4.11	TIMx prescaler (TIMx_PSC)	638
18.4.12	TIMx auto-reload register (TIMx_ARR)	638
18.4.13	TIMx capture/compare register 1 (TIMx_CCR1)	639
18.4.14	TIMx capture/compare register 2 (TIMx_CCR2)	639
18.4.15	TIMx capture/compare register 3 (TIMx_CCR3)	640
18.4.16	TIMx capture/compare register 4 (TIMx_CCR4)	640
18.4.17	TIMx DMA control register (TIMx_DCR)	641
18.4.18	TIMx DMA address for full transfer (TIMx_DMAR)	641
18.4.19	TIM2 option register (TIM2_OR)	642
18.4.20	TIM5 option register (TIM5_OR)	643
18.4.21	TIMx register map	644
19	General-purpose timers (TIM9 to TIM14)	646
19.1	TIM9 to TIM14 introduction	646
19.2	TIM9 to TIM14 main features	646
19.2.1	TIM9/TIM12 main features	646
19.2.2	TIM10/TIM11 and TIM13/TIM14 main features	647
19.3	TIM9 to TIM14 functional description	649
19.3.1	Time-base unit	649
19.3.2	Counter modes	651
19.3.3	Clock selection	654
19.3.4	Capture/compare channels	656
19.3.5	Input capture mode	657
19.3.6	PWM input mode (only for TIM9/12)	659
19.3.7	Forced output mode	660

19.3.8	Output compare mode	660
19.3.9	PWM mode	661
19.3.10	One-pulse mode	662
19.3.11	TIM9/12 external trigger synchronization	664
19.3.12	Timer synchronization (TIM9/12)	667
19.3.13	Debug mode	667
19.4	TIM9 and TIM12 registers	668
19.4.1	TIM9/12 control register 1 (TIMx_CR1)	668
19.4.2	TIM9/12 slave mode control register (TIMx_SMCR)	669
19.4.3	TIM9/12 Interrupt enable register (TIMx_DIER)	670
19.4.4	TIM9/12 status register (TIMx_SR)	672
19.4.5	TIM9/12 event generation register (TIMx_EGR)	673
19.4.6	TIM9/12 capture/compare mode register 1 (TIMx_CCMR1)	674
19.4.7	TIM9/12 capture/compare enable register (TIMx_CCER)	677
19.4.8	TIM9/12 counter (TIMx_CNT)	678
19.4.9	TIM9/12 prescaler (TIMx_PSC)	678
19.4.10	TIM9/12 auto-reload register (TIMx_ARR)	678
19.4.11	TIM9/12 capture/compare register 1 (TIMx_CCR1)	679
19.4.12	TIM9/12 capture/compare register 2 (TIMx_CCR2)	679
19.4.13	TIM9/12 register map	680
19.5	TIM10/11/13/14 registers	682
19.5.1	TIM10/11/13/14 control register 1 (TIMx_CR1)	682
19.5.2	TIM10/11/13/14 Interrupt enable register (TIMx_DIER)	683
19.5.3	TIM10/11/13/14 status register (TIMx_SR)	683
19.5.4	TIM10/11/13/14 event generation register (TIMx_EGR)	684
19.5.5	TIM10/11/13/14 capture/compare mode register 1 (TIMx_CCMR1)	684
19.5.6	TIM10/11/13/14 capture/compare enable register (TIMx_CCER)	687
19.5.7	TIM10/11/13/14 counter (TIMx_CNT)	688
19.5.8	TIM10/11/13/14 prescaler (TIMx_PSC)	688
19.5.9	TIM10/11/13/14 auto-reload register (TIMx_ARR)	688
19.5.10	TIM10/11/13/14 capture/compare register 1 (TIMx_CCR1)	689
19.5.11	TIM11 option register 1 (TIM11_OR)	689
19.5.12	TIM10/11/13/14 register map	690
20	Basic timers (TIM6 and TIM7)	692
20.1	TIM6 and TIM7 introduction	692
20.2	TIM6 and TIM7 main features	692

20.3	TIM6 and TIM7 functional description	693
20.3.1	Time-base unit	693
20.3.2	Counting mode	695
20.3.3	Clock source	697
20.3.4	Debug mode	698
20.4	TIM6 and TIM7 registers	698
20.4.1	TIM6 and TIM7 control register 1 (TIMx_CR1)	698
20.4.2	TIM6 and TIM7 control register 2 (TIMx_CR2)	700
20.4.3	TIM6 and TIM7 DMA/Interrupt enable register (TIMx_DIER)	700
20.4.4	TIM6 and TIM7 status register (TIMx_SR)	701
20.4.5	TIM6 and TIM7 event generation register (TIMx_EGR)	701
20.4.6	TIM6 and TIM7 counter (TIMx_CNT)	701
20.4.7	TIM6 and TIM7 prescaler (TIMx_PSC)	702
20.4.8	TIM6 and TIM7 auto-reload register (TIMx_ARR)	702
20.4.9	TIM6 and TIM7 register map	703
21	Independent watchdog (IWDG)	704
21.1	IWDG introduction	704
21.2	IWDG main features	704
21.3	IWDG functional description	704
21.3.1	Hardware watchdog	704
21.3.2	Register access protection	704
21.3.3	Debug mode	705
21.4	IWDG registers	706
21.4.1	Key register (IWDG_KR)	706
21.4.2	Prescaler register (IWDG_PR)	706
21.4.3	Reload register (IWDG_RLR)	707
21.4.4	Status register (IWDG_SR)	707
21.4.5	IWDG register map	708
22	Window watchdog (WWDG)	709
22.1	WWDG introduction	709
22.2	WWDG main features	709
22.3	WWDG functional description	709
22.4	How to program the watchdog timeout	711
22.5	Debug mode	712

22.6	WWDG registers	713
22.6.1	Control register (WWDG_CR)	713
22.6.2	Configuration register (WWDG_CFR)	714
22.6.3	Status register (WWDG_SR)	714
22.6.4	WWDG register map	715
23	Cryptographic processor (CRYP)	716
23.1	CRYP introduction	716
23.2	CRYP main features	716
23.3	CRYP functional description	718
23.3.1	DES/TDES cryptographic core	719
23.3.2	AES cryptographic core	724
23.3.3	Data type	735
23.3.4	Initialization vectors - CRYP_IV0...1(L/R)	738
23.3.5	CRYP busy state	739
23.3.6	Procedure to perform an encryption or a decryption	740
23.3.7	Context swapping	741
23.4	CRYP interrupts	743
23.5	CRYP DMA interface	744
23.6	CRYP registers	744
23.6.1	CRYP control register (CRYP_CR) for STM32F415/417xx	744
23.6.2	CRYP control register (CRYP_CR) for STM32F415/417xx	746
23.6.3	CRYP status register (CRYP_SR)	749
23.6.4	CRYP data input register (CRYP_DIN)	750
23.6.5	CRYP data output register (CRYP_DOUT)	751
23.6.6	CRYP DMA control register (CRYP_DMACR)	752
23.6.7	CRYP interrupt mask set/clear register (CRYP_IMSCR)	752
23.6.8	CRYP raw interrupt status register (CRYP_RISR)	753
23.6.9	CRYP masked interrupt status register (CRYP_MISR)	753
23.6.10	CRYP key registers (CRYP_K0...3(L/R)R)	754
23.6.11	CRYP initialization vector registers (CRYP_IV0...1(L/R)R)	756
23.6.12	CRYP context swap registers (CRYP_CSGCMCCM0..7R and CRYP_CSGCM0..7R) for STM32F42xxx and STM32F43xxx	758
23.6.13	CRYP register map	759
24	Random number generator (RNG)	763
24.1	RNG introduction	763

24.2	RNG main features	763
24.3	RNG functional description	763
24.3.1	Operation	764
24.3.2	Error management	764
24.4	RNG registers	764
24.4.1	RNG control register (RNG_CR)	765
24.4.2	RNG status register (RNG_SR)	765
24.4.3	RNG data register (RNG_DR)	766
24.4.4	RNG register map	767
25	Hash processor (HASH)	768
25.1	HASH introduction	768
25.2	HASH main features	768
25.3	HASH functional description	769
25.3.1	Duration of the processing	771
25.3.2	Data type	771
25.3.3	Message digest computing	773
25.3.4	Message padding	774
25.3.5	Hash operation	775
25.3.6	HMAC operation	775
25.3.7	Context swapping	776
25.3.8	HASH interrupt	778
25.4	HASH registers	778
25.4.1	HASH control register (HASH_CR) for STM32F415/417xx	778
25.4.2	HASH control register (HASH_CR) for STM32F43xxx	781
25.4.3	HASH data input register (HASH_DIN)	784
25.4.4	HASH start register (HASH_STR)	785
25.4.5	HASH digest registers (HASH_HR0..4/5/6/7)	786
25.4.6	HASH interrupt enable register (HASH_IMR)	788
25.4.7	HASH status register (HASH_SR)	789
25.4.8	HASH context swap registers (HASH_CSRx)	790
25.4.9	HASH register map	791
26	Real-time clock (RTC)	794
26.1	Introduction	794
26.2	RTC main features	795

26.3	RTC functional description	797
26.3.1	Clock and prescalers	797
26.3.2	Real-time clock and calendar	798
26.3.3	Programmable alarms	798
26.3.4	Periodic auto-wakeup	798
26.3.5	RTC initialization and configuration	799
26.3.6	Reading the calendar	801
26.3.7	Resetting the RTC	802
26.3.8	RTC synchronization	802
26.3.9	RTC reference clock detection	803
26.3.10	RTC coarse digital calibration	804
26.3.11	RTC smooth digital calibration	805
26.3.12	Timestamp function	807
26.3.13	Tamper detection	807
26.3.14	Calibration clock output	809
26.3.15	Alarm output	809
26.4	RTC and low-power modes	810
26.5	RTC interrupts	810
26.6	RTC registers	812
26.6.1	RTC time register (RTC_TR)	812
26.6.2	RTC date register (RTC_DR)	813
26.6.3	RTC control register (RTC_CR)	814
26.6.4	RTC initialization and status register (RTC_ISR)	816
26.6.5	RTC prescaler register (RTC_PRER)	819
26.6.6	RTC wake-up timer register (RTC_WUTR)	819
26.6.7	RTC calibration register (RTC_CALIBR)	820
26.6.8	RTC alarm A register (RTC_ALRMAR)	821
26.6.9	RTC alarm B register (RTC_ALRMBR)	822
26.6.10	RTC write protection register (RTC_WPR)	823
26.6.11	RTC sub second register (RTC_SSR)	823
26.6.12	RTC shift control register (RTC_SHIFTR)	824
26.6.13	RTC time stamp time register (RTC_TSTR)	824
26.6.14	RTC time stamp date register (RTC_TSDR)	825
26.6.15	RTC timestamp sub second register (RTC_TSSSR)	826
26.6.16	RTC calibration register (RTC_CALR)	826
26.6.17	RTC tamper and alternate function configuration register (RTC_TAFCR)	828

26.6.18	RTC alarm A sub second register (RTC_ALRMASR)	830
26.6.19	RTC alarm B sub second register (RTC_ALRMBSSR)	831
26.6.20	RTC backup registers (RTC_BKPxR)	832
26.6.21	RTC register map	832
27	Inter-integrated circuit (I2C) interface	835
27.1	I ² C introduction	835
27.2	I ² C main features	835
27.3	I ² C functional description	836
27.3.1	Mode selection	836
27.3.2	I2C target mode	839
27.3.3	I2C controller mode	841
27.3.4	Error conditions	846
27.3.5	Programmable noise filter	847
27.3.6	SDA/SCL line control	848
27.3.7	SMBus	848
27.3.8	DMA requests	851
27.3.9	Packet error checking	852
27.4	I ² C interrupts	853
27.5	I ² C debug mode	855
27.6	I ² C registers	855
27.6.1	I ² C Control register 1 (I2C_CR1)	855
27.6.2	I ² C Control register 2 (I2C_CR2)	857
27.6.3	I ² C Own address register 1 (I2C_OAR1)	859
27.6.4	I ² C Own address register 2 (I2C_OAR2)	859
27.6.5	I ² C Data register (I2C_DR)	860
27.6.6	I ² C Status register 1 (I2C_SR1)	860
27.6.7	I ² C Status register 2 (I2C_SR2)	863
27.6.8	I ² C Clock control register (I2C_CCR)	865
27.6.9	I ² C TRISE register (I2C_TRISE)	866
27.6.10	I ² C FLTR register (I2C_FLTR)	866
27.6.11	I2C register map	867
28	Serial peripheral interface (SPI)	868
28.1	SPI introduction	868
28.2	SPI and I ² S main features	869

28.2.1	SPI features	869
28.2.2	I ² S features	869
28.3	SPI functional description	871
28.3.1	General description	871
28.3.2	Configuring the SPI in slave mode	875
28.3.3	Configuring the SPI in master mode	877
28.3.4	Configuring the SPI for half-duplex communication	879
28.3.5	Data transmission and reception procedures	880
28.3.6	CRC calculation	886
28.3.7	Status flags	887
28.3.8	Disabling the SPI	888
28.3.9	SPI communication using DMA (direct memory addressing)	889
28.3.10	Error flags	891
28.3.11	SPI interrupts	893
28.4	I ² S functional description	894
28.4.1	I ² S general description	894
28.4.2	I2S full duplex	895
28.4.3	Supported audio protocols	896
28.4.4	Clock generator	902
28.4.5	I ² S master mode	904
28.4.6	I ² S slave mode	906
28.4.7	Status flags	908
28.4.8	Error flags	909
28.4.9	I ² S interrupts	910
28.4.10	DMA features	910
28.5	SPI and I ² S registers	911
28.5.1	SPI control register 1 (SPI_CR1) (not used in I ² S mode)	911
28.5.2	SPI control register 2 (SPI_CR2)	913
28.5.3	SPI status register (SPI_SR)	914
28.5.4	SPI data register (SPI_DR)	915
28.5.5	SPI CRC polynomial register (SPI_CRCPR) (not used in I ² S mode)	916
28.5.6	SPI RX CRC register (SPI_RXCR) (not used in I ² S mode)	916
28.5.7	SPI TX CRC register (SPI_TXCR) (not used in I ² S mode)	916
28.5.8	SPI_I ² S configuration register (SPI_I2SCFGR)	917
28.5.9	SPI_I ² S prescaler register (SPI_I2SPR)	918
28.5.10	SPI register map	920

29	Serial audio interface (SAI)	921
29.1	Introduction	921
29.2	Main features	922
29.3	Functional block diagram	923
29.4	Main SAI modes	924
29.5	SAI synchronization mode	925
29.6	Audio data size	925
29.7	Frame synchronization	925
29.7.1	Frame length	926
29.7.2	Frame synchronization polarity	926
29.7.3	Frame synchronization active level length	927
29.7.4	Frame synchronization offset	927
29.7.5	FS signal role	927
29.8	Slot configuration	928
29.9	SAI clock generator	930
29.10	Internal FIFOs	931
29.11	AC'97 link controller	934
29.12	Specific features	935
29.12.1	Mute mode	935
29.12.2	MONO/STEREO function	935
29.12.3	Companding mode	936
29.12.4	Output data line management on an inactive slot	937
29.13	Error flags	939
29.13.1	FIFO overrun/underrun (OVRUDR)	939
29.13.2	Anticipated frame synchronisation detection (AFSDET)	941
29.13.3	Late frame synchronization detection	941
29.13.4	Codec not ready (CNRDY AC'97)	942
29.13.5	Wrong clock configuration in master mode (with NODIV = 0)	942
29.14	Interrupt sources	943
29.15	Disabling the SAI	943
29.16	SAI DMA interface	944
29.17	SAI registers	945
29.17.1	SAI x configuration register 1 (SAI_xCR1) where x is A or B	945
29.17.2	SAI x configuration register 2 (SAI_xCR2) where x is A or B	948
29.17.3	SAI x frame configuration register (SAI_XFRCR) where x is A or B	950

29.17.4	SAI x slot register (SAI_xSLOTR) where x is A or B	952
29.17.5	SAI x interrupt mask register (SAI_xIM) where x is A or B	953
29.17.6	SAI x status register (SAI_xSR) where x is A or B	955
29.17.7	SAI x clear flag register (SAI_xCLRFR) where X is A or B	957
29.17.8	SAI x data register (SAI_xDR) where x is A or B	958
29.17.9	SAI register map	958
30	Universal synchronous asynchronous receiver transmitter (USART)	960
30.1	USART introduction	960
30.2	USART main features	960
30.3	USART functional description	961
30.3.1	USART character description	964
30.3.2	Transmitter	965
30.3.3	Receiver	968
30.3.4	Fractional baud rate generation	973
30.3.5	USART receiver tolerance to clock deviation	983
30.3.6	Multiprocessor communication	984
30.3.7	Parity control	986
30.3.8	LIN (local interconnection network) mode	987
30.3.9	USART synchronous mode	989
30.3.10	Single-wire half-duplex communication	991
30.3.11	Smartcard	992
30.3.12	IrDA SIR ENDEC block	994
30.3.13	Continuous communication using DMA	996
30.3.14	Hardware flow control	998
30.4	USART interrupts	1001
30.5	USART mode configuration	1002
30.6	USART registers	1002
30.6.1	Status register (USART_SR)	1002
30.6.2	Data register (USART_DR)	1005
30.6.3	Baud rate register (USART_BRR)	1005
30.6.4	Control register 1 (USART_CR1)	1005
30.6.5	Control register 2 (USART_CR2)	1008
30.6.6	Control register 3 (USART_CR3)	1009
30.6.7	Guard time and prescaler register (USART_GTPR)	1012
30.6.8	USART register map	1013

31	Secure digital input/output interface (SDIO)	1014
31.1	SDIO main features	1014
31.2	SDIO bus topology	1015
31.3	SDIO functional description	1017
31.3.1	SDIO adapter	1018
31.3.2	SDIO APB2 interface	1028
31.4	Card functional description	1029
31.4.1	Card identification mode	1029
31.4.2	Card reset	1029
31.4.3	Operating voltage range validation	1029
31.4.4	Card identification process	1030
31.4.5	Block write	1031
31.4.6	Block read	1031
31.4.7	Stream access, stream write and stream read (MultiMediaCard only) ... 1032	
31.4.8	Erase: group erase and sector erase	1033
31.4.9	Wide bus selection or deselection	1034
31.4.10	Protection management	1034
31.4.11	Card status register	1037
31.4.12	SD status register	1040
31.4.13	SD I/O mode	1044
31.4.14	Commands and responses	1045
31.5	Response formats	1048
31.5.1	R1 (normal response command)	1049
31.5.2	R1b	1049
31.5.3	R2 (CID, CSD register)	1049
31.5.4	R3 (OCR register)	1049
31.5.5	R4 (Fast I/O)	1050
31.5.6	R4b	1050
31.5.7	R5 (interrupt request)	1051
31.5.8	R6	1051
31.6	SDIO I/O card-specific operations	1052
31.6.1	SDIO I/O read wait operation by SDIO_D2 signaling	1052
31.6.2	SDIO read wait operation by stopping SDIO_CK	1052
31.6.3	SDIO suspend/resume operation	1053
31.6.4	SDIO interrupts	1053

31.7	CE-ATA specific operations	1053
31.7.1	Command completion signal disable	1053
31.7.2	Command completion signal enable	1053
31.7.3	CE-ATA interrupt	1054
31.7.4	Aborting CMD61	1054
31.8	HW flow control	1054
31.9	SDIO registers	1054
31.9.1	SDIO power control register (SDIO_POWER)	1054
31.9.2	SDI clock control register (SDIO_CLKCR)	1055
31.9.3	SDIO argument register (SDIO_ARG)	1056
31.9.4	SDIO command register (SDIO_CMD)	1056
31.9.5	SDIO command response register (SDIO_RESPCMD)	1057
31.9.6	SDIO response 1..4 register (SDIO_RESPx)	1058
31.9.7	SDIO data timer register (SDIO_DTIMER)	1058
31.9.8	SDIO data length register (SDIO_DLEN)	1059
31.9.9	SDIO data control register (SDIO_DCTRL)	1060
31.9.10	SDIO data counter register (SDIO_DCOUNT)	1061
31.9.11	SDIO status register (SDIO_STA)	1061
31.9.12	SDIO interrupt clear register (SDIO_ICR)	1063
31.9.13	SDIO mask register (SDIO_MASK)	1064
31.9.14	SDIO FIFO counter register (SDIO_FIFOCNT)	1067
31.9.15	SDIO data FIFO register (SDIO_FIFO)	1067
31.9.16	SDIO register map	1068
32	Controller area network (bxCAN)	1069
32.1	bxCAN introduction	1069
32.2	bxCAN main features	1069
32.3	bxCAN general description	1070
32.3.1	CAN 2.0B active core	1070
32.3.2	Control, status and configuration registers	1071
32.3.3	Tx mailboxes	1071
32.3.4	Acceptance filters	1071
32.4	bxCAN operating modes	1072
32.4.1	Initialization mode	1073
32.4.2	Normal mode	1073
32.4.3	Sleep mode (low-power)	1073

32.5	Test mode	1074
32.5.1	Silent mode	1074
32.5.2	Loop back mode	1075
32.5.3	Loop back combined with silent mode	1075
32.6	Debug mode	1076
32.7	bxCAN functional description	1076
32.7.1	Transmission handling	1076
32.7.2	Time triggered communication mode	1078
32.7.3	Reception handling	1078
32.7.4	Identifier filtering	1080
32.7.5	Message storage	1084
32.7.6	Error management	1086
32.7.7	Bit timing	1086
32.8	bxCAN interrupts	1088
32.9	CAN registers	1090
32.9.1	Register access protection	1090
32.9.2	CAN control and status registers	1090
32.9.3	CAN mailbox registers	1099
32.9.4	CAN filter registers	1104
32.9.5	bxCAN register map	1109
33	Ethernet (ETH): media access control (MAC) with DMA controller	1113
33.1	Ethernet introduction	1113
33.2	Ethernet main features	1113
33.2.1	MAC core features	1114
33.2.2	DMA features	1115
33.2.3	PTP features	1115
33.3	Ethernet pins	1116
33.4	Ethernet functional description: SMI, MII and RMII	1117
33.4.1	Station management interface: SMI	1117
33.4.2	Media-independent interface: MII	1120
33.4.3	Reduced media-independent interface: RMII	1122
33.4.4	II/RMII selection	1123
33.5	Ethernet functional description: MAC 802.3	1124
33.5.1	MAC 802.3 frame format	1125

33.5.2	MAC frame transmission	1128
33.5.3	MAC frame reception	1135
33.5.4	MAC interrupts	1140
33.5.5	MAC filtering	1141
33.5.6	MAC loopback mode	1144
33.5.7	MAC management counters: MMC	1144
33.5.8	Power management: PMT	1145
33.5.9	Precision time protocol (IEEE1588 PTP)	1148
33.6	Ethernet functional description: DMA controller operation	1154
33.6.1	Initialization of a transfer using DMA	1155
33.6.2	Host bus burst access	1155
33.6.3	Host data buffer alignment	1156
33.6.4	Buffer size calculations	1156
33.6.5	DMA arbiter	1157
33.6.6	Error response to DMA	1157
33.6.7	Tx DMA configuration	1157
33.6.8	Rx DMA configuration	1169
33.6.9	DMA interrupts	1179
33.7	Ethernet interrupts	1180
33.8	Ethernet register descriptions	1181
33.8.1	MAC register description	1181
33.8.2	MMC register description	1198
33.8.3	IEEE 1588 time stamp registers	1204
33.8.4	DMA register description	1210
33.8.5	Ethernet register maps	1224
34	USB on-the-go full-speed (OTG_FS)	1228
34.1	OTG_FS introduction	1228
34.2	OTG_FS main features	1229
34.2.1	General features	1229
34.2.2	Host-mode features	1230
34.2.3	Peripheral-mode features	1230
34.3	OTG_FS functional description	1231
34.3.1	OTG pins	1231
34.3.2	OTG full-speed core	1231
34.3.3	Full-speed OTG PHY	1232

34.4	OTG dual role device (DRD)	1233
34.4.1	ID line detection	1233
34.4.2	HNP dual role device	1233
34.4.3	SRP dual role device	1234
34.5	USB peripheral	1234
34.5.1	SRP-capable peripheral	1235
34.5.2	Peripheral states	1235
34.5.3	Peripheral endpoints	1236
34.6	USB host	1238
34.6.1	SRP-capable host	1239
34.6.2	USB host states	1239
34.6.3	Host channels	1241
34.6.4	Host scheduler	1242
34.7	SOF trigger	1243
34.7.1	Host SOFs	1243
34.7.2	Peripheral SOFs	1244
34.8	OTG low-power modes	1244
34.9	Dynamic update of the OTG_FS_HFIR register	1245
34.10	USB data FIFOs	1246
34.11	Peripheral FIFO architecture	1247
34.11.1	Peripheral Rx FIFO	1247
34.11.2	Peripheral Tx FIFOs	1248
34.12	Host FIFO architecture	1248
34.12.1	Host Rx FIFO	1248
34.12.2	Host Tx FIFOs	1249
34.13	FIFO RAM allocation	1249
34.13.1	Device mode	1249
34.13.2	Host mode	1250
34.14	USB system performance	1250
34.15	OTG_FS interrupts	1251
34.16	OTG_FS control and status registers	1253
34.16.1	CSR memory map	1254
34.16.2	OTG_FS global registers	1259
34.16.3	Host-mode registers	1280
34.16.4	Device-mode registers	1290

34.16.5	OTG_FS power and clock gating control register (OTG_FS_PCGCCTL)	1313
34.16.6	OTG_FS register map	1314
34.17	OTG_FS programming model	1323
34.17.1	Core initialization	1323
34.17.2	Host initialization	1324
34.17.3	Device initialization	1324
34.17.4	Host programming model	1325
34.17.5	Device programming model	1341
34.17.6	Operational model	1343
34.17.7	Worst case response time	1361
34.17.8	OTG programming model	1362
35	USB on-the-go high-speed (OTG_HS)	1369
35.1	OTG_HS introduction	1369
35.2	OTG_HS main features	1369
35.2.1	General features	1370
35.2.2	Host-mode features	1371
35.2.3	Peripheral-mode features	1371
35.3	OTG_HS functional description	1371
35.3.1	OTG pins	1372
35.3.2	High-speed OTG PHY	1372
35.3.3	Embedded Full-speed OTG PHY	1373
35.4	OTG dual-role device	1373
35.4.1	ID line detection	1373
35.4.2	HNP dual role device	1373
35.4.3	SRP dual-role device	1374
35.5	USB functional description in peripheral mode	1374
35.5.1	SRP-capable peripheral	1374
35.5.2	Peripheral states	1375
35.5.3	Peripheral endpoints	1376
35.6	USB functional description on host mode	1379
35.6.1	SRP-capable host	1379
35.6.2	USB host states	1380
35.6.3	Host channels	1381
35.6.4	Host scheduler	1383

35.7	SOF trigger	1384
35.7.1	Host SOFs	1384
35.7.2	Peripheral SOFs	1384
35.8	OTG_HS low-power modes	1385
35.9	Dynamic update of the OTG_HS_HFIR register	1386
35.10	FIFO RAM allocation	1387
35.10.1	Peripheral mode	1387
35.10.2	Host mode	1387
35.11	OTG_HS interrupts	1388
35.12	OTG_HS control and status registers	1390
35.12.1	CSR memory map	1390
35.12.2	OTG_HS global registers	1395
35.12.3	Host-mode registers	1418
35.12.4	Device-mode registers	1431
35.12.5	OTG_HS power and clock gating control register (OTG_HS_PCGCTL)	1460
35.12.6	OTG_HS register map	1460
35.13	OTG_HS programming model	1475
35.13.1	Core initialization	1475
35.13.2	Host initialization	1476
35.13.3	Device initialization	1477
35.13.4	DMA mode	1477
35.13.5	Host programming model	1477
35.13.6	Device programming model	1503
35.13.7	Operational model	1505
35.13.8	Worst case response time	1524
35.13.9	OTG programming model	1526
36	Flexible static memory controller (FSMC)	1532
36.1	FSMC main features	1532
36.2	Block diagram	1533
36.3	AHB interface	1533
36.3.1	Supported memories and transactions	1534
36.4	External device address mapping	1535
36.4.1	NOR/PSRAM address mapping	1535
36.4.2	NAND/PC Card address mapping	1536

36.5	NOR flash/PSRAM controller	1537
36.5.1	External memory interface signals	1538
36.5.2	Supported memories and transactions	1540
36.5.3	General timing rules	1541
36.5.4	NOR flash/PSRAM controller asynchronous transactions	1542
36.5.5	Synchronous transactions	1559
36.5.6	NOR/PSRAM control registers	1565
36.6	NAND flash/PC Card controller	1572
36.6.1	External memory interface signals	1573
36.6.2	NAND flash / PC Card supported memories and transactions	1575
36.6.3	Timing diagrams for NAND and PC Card	1575
36.6.4	NAND flash operations	1576
36.6.5	NAND flash prewait functionality	1577
36.6.6	Computation of the error correction code (ECC) in NAND flash memory	1578
36.6.7	PC Card/CompactFlash operations	1579
36.6.8	NAND flash/PC Card control registers	1581
36.6.9	FSMC register map	1588
37	Flexible memory controller (FMC)	1590
37.1	FMC main features	1590
37.2	Block diagram	1591
37.3	AHB interface	1592
37.3.1	Supported memories and transactions	1593
37.4	External device address mapping	1594
37.4.1	NOR/PSRAM address mapping	1595
37.4.2	NAND Flash memory/PC Card address mapping	1596
37.4.3	SDRAM address mapping	1597
37.5	NOR Flash/PSRAM controller	1600
37.5.1	External memory interface signals	1601
37.5.2	Supported memories and transactions	1603
37.5.3	General timing rules	1605
37.5.4	NOR Flash/PSRAM controller asynchronous transactions	1605
37.5.5	Synchronous transactions	1622
37.5.6	NOR/PSRAM controller registers	1628
37.6	NAND Flash/PC Card controller	1635
37.6.1	External memory interface signals	1636

37.6.2	NAND Flash / PC Card supported memories and transactions	1638
37.6.3	Timing diagrams for NAND Flash memory and PC Card	1638
37.6.4	NAND Flash operations	1639
37.6.5	NAND Flash prewait functionality	1640
37.6.6	Computation of the error correction code (ECC) in NAND Flash memory	1641
37.6.7	PC Card/CompactFlash operations	1642
37.6.8	NAND Flash/PC Card controller registers	1644
37.7	SDRAM controller	1651
37.7.1	SDRAM controller main features	1651
37.7.2	SDRAM External memory interface signals	1651
37.7.3	SDRAM controller functional description	1652
37.7.4	Low power modes	1659
37.7.5	SDRAM controller registers	1662
37.8	FMC register map	1668
38	Debug support (DBG)	1671
38.1	Overview	1671
38.2	Reference Arm® documentation	1672
38.3	SWJ debug port (serial wire and JTAG)	1672
38.3.1	Mechanism to select the JTAG-DP or the SW-DP	1673
38.4	Pinout and debug port pins	1673
38.4.1	SWJ debug port pins	1674
38.4.2	Flexible SWJ-DP pin assignment	1674
38.4.3	Internal pull-up and pull-down on JTAG pins	1674
38.4.4	Using serial wire and releasing the unused debug pins as GPIOs . .	1675
38.5	STM32F4xx JTAG TAP connection	1675
38.6	ID codes and locking mechanism	1676
38.6.1	MCU device ID code	1676
38.6.2	Boundary scan TAP	1677
38.6.3	Cortex®-M4 with FPU TAP	1678
38.6.4	Cortex®-M4 with FPU JEDEC-106 ID code	1678
38.7	JTAG debug port	1678
38.8	SW debug port	1680
38.8.1	SW protocol introduction	1680
38.8.2	SW protocol sequence	1680

38.8.3	SW-DP state machine (reset, idle states, ID code)	1681
38.8.4	DP and AP read/write accesses	1681
38.8.5	SW-DP registers	1682
38.8.6	SW-AP registers	1682
38.9	AHB-AP (AHB access port) - valid for both JTAG-DP and SW-DP	1683
38.10	Core debug	1683
38.11	Capability of the debugger host to connect under system reset	1684
38.12	FPB (Flash patch breakpoint)	1685
38.13	DWT (data watchpoint trigger)	1686
38.14	ITM (instrumentation trace macrocell)	1686
38.14.1	General description	1686
38.14.2	Time stamp packets, synchronization and overflow packets	1686
38.15	ETM (Embedded Trace Macrocell™)	1688
38.15.1	ETM general description	1688
38.15.2	ETM signal protocol and packet types	1688
38.15.3	Main ETM registers	1689
38.15.4	ETM configuration example	1689
38.16	MCU debug component (DBGMCU)	1689
38.16.1	Debug support for low-power modes	1689
38.16.2	Debug support for timers, watchdog, bxCAN and I ² C	1690
38.16.3	Debug MCU configuration register	1690
38.16.4	Debug MCU APB1 freeze register (DBGMCU_APB1_FZ)	1692
38.16.5	Debug MCU APB2 Freeze register (DBGMCU_APB2_FZ)	1693
38.17	TPIU (trace port interface unit)	1694
38.17.1	Introduction	1694
38.17.2	TRACE pin assignment	1695
38.17.3	TPUI formatter	1696
38.17.4	TPUI frame synchronization packets	1697
38.17.5	Transmission of the synchronization frame packet	1697
38.17.6	Synchronous mode	1697
38.17.7	Asynchronous mode	1698
38.17.8	TRACECLKIN connection inside the STM32F4xx	1698
38.17.9	TPIU registers	1698
38.17.10	Example of configuration	1699
38.18	DBG register map	1700

39	Device electronic signature	1701
	39.1 Unique device ID register (96 bits)	1701
	39.2 Flash size	1702
40	Important security notice	1703
41	Revision history	1704